

DURHAM 2 N, NH, USAWMO: **726090**Lat: **43.1716N**Lon: **70.9277W**Elev: **119**StdP: **14.63**Time Zone: **-5.00 (NAE)**Period: **02-19**WBAN: **54794****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	4.9	-13.3	2.7	5.7	-8.7	3.5	10.5	11.8	31.4	10.4	28.1	2.7	N/A	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	87.7	72.1	84.8	70.5	82.0	68.9	74.9	84.0	73.2	81.5	71.6	78.7	4.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	118.3	79.2	70.5	112.9	77.5	69.2	107.6	75.9	38.3	83.8	36.8	81.6	35.4	78.7	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.8	7.8	-8.1	92.8	5.0	2.0	-11.7	94.2	-14.6	95.4	-17.5	96.5	-21.1	98.0		
(5)			-9.4	77.5	3.6	1.1	-11.9	78.3	-14.0	78.9	-16.0	79.5	-18.6	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	24.0	26.3	33.8	45.3	55.3	63.8	70.2	68.5	61.5	50.0	39.8	29.7		
	DBStd	17.81	10.55	9.56	9.20	8.18	7.49	6.65	5.00	5.14	6.79	7.49	8.17	8.81		
	HDD50	3241	808	665	509	189	28	1	0	0	1	93	318	629		
	HDD65	6829	1272	1085	968	594	316	100	11	23	146	466	755	1093		
	CDD50	2315	1	0	6	46	191	415	626	574	348	94	13	1		
	CDD65	431	0	0	0	2	15	65	173	132	42	2	0	0		
	CDH74	3801	0	0	7	64	235	605	1456	1058	356	20	0	0		
	CDH80	1019	0	0	1	17	64	163	420	266	87	1	0	0		
	WSAvg	3.1	3.4	3.7	4.0	4.2	3.6	2.9	2.3	2.0	2.1	2.8	3.0	3.2		
	PrecAvg	40.6	3.0	2.9	3.1	3.7	3.3	3.2	3.0	3.1	3.3	3.9	4.2	3.8		
(15) Precipitation	PrecMax	58.8	7.8	10.3	9.2	11.5	9.3	11.3	7.9	8.8	9.3	13.9	11.3	9.3		
	PrecMin	24.6	0.5	0.0	0.9	0.6	0.7	0.8	0.6	0.8	0.3	0.2	0.6	0.5		
	PrecStd	7.4	1.7	1.6	1.5	2.1	1.8	2.0	1.7	1.6	1.8	2.8	2.1	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	57.4	70.3	82.1	87.0	89.8	91.2	90.0	88.2	76.9	67.7	57.4		
		MCWB	54.7	48.2	56.4	61.5	66.6	72.2	74.3	74.5	73.6	65.9	58.1	53.3		
	2%	DB	48.3	49.7	59.0	72.9	80.6	84.6	87.5	86.2	82.2	71.5	62.1	50.8		
		MCWB	43.0	42.1	50.0	59.3	63.9	70.3	72.7	71.9	70.2	62.7	54.4	47.1		
	5%	DB	42.5	44.3	52.1	66.1	75.1	80.7	84.8	82.9	77.9	66.4	57.6	46.8		
		MCWB	37.9	38.0	43.3	55.2	60.9	66.8	71.0	69.9	67.9	60.4	50.4	42.8		
	10%	DB	38.2	39.6	46.8	60.5	70.0	77.1	81.8	79.9	74.0	62.3	53.4	42.3		
		MCWB	33.9	34.5	40.0	49.1	59.4	64.3	69.9	68.0	66.2	56.7	47.9	38.7		
	0.4%	WB	55.1	53.6	59.0	64.5	70.6	74.4	77.3	76.0	75.3	68.7	60.5	53.6		
		MCDB	56.8	57.6	72.2	71.4	82.9	84.2	87.9	85.7	84.2	72.8	65.2	54.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.2	45.0	53.1	60.4	67.6	72.2	75.4	73.7	72.6	64.4	56.1	48.3		
		MCDB	45.9	50.5	60.7	70.1	76.8	81.7	84.7	81.5	80.0	68.3	59.8	50.7		
	5%	WB	38.6	39.6	44.8	55.9	64.4	69.6	73.8	72.0	70.5	61.7	51.9	44.4		
		MCDB	42.0	44.8	49.8	63.1	72.6	77.0	82.2	79.1	76.6	66.0	56.1	46.8		
	10%	WB	34.5	36.3	40.3	50.6	61.3	67.2	72.2	70.4	68.4	58.5	48.2	39.8		
		MCDB	37.8	40.6	45.6	57.8	68.6	74.0	79.5	76.3	73.3	62.4	52.6	42.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	17.7	17.7	19.9	19.7	19.3	19.7	20.4	20.6	18.5	16.9	15.0		
		MCDBR	20.1	23.2	25.3	31.1	29.1	25.0	23.7	23.9	24.7	23.7	23.4	20.1		
		MCWBR	18.3	16.9	16.8	17.2	13.9	12.3	11.0	11.5	13.2	14.8	17.2	16.2		
		MCDBR	20.2	23.4	24.9	24.7	24.5	21.7	20.6	19.5	21.1	19.6	20.9	18.4		
	5% WB	MCWBR	18.6	18.0	17.6	16.5	14.0	11.9	10.9	10.6	12.6	14.3	17.4	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.296	0.320	0.356	0.393	0.440	0.459	0.426	0.373	0.333	0.312	0.291		
	taud		2.467	2.449	2.468	2.418	2.353	2.253	2.206	2.285	2.416	2.528	2.547	2.534		
	Ebn at Noon		272	288	293	289	279	266	259	264	271	271	261	261		
	Edh at Noon		23	28	31	35	38	43	44	40	32	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		541	816	1155	1460	1626	1751	1842	1670	1345	869	581	436		
	RadStd		36	64	74	158	186	155	121	82	82	79	52	47		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+59

Nomenclature: See separate page